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[22TCSE219]

FIFTH SEMESTER B.Tech EXAMINATION, SEPTEMBER-2024

CLOUD SECURITY

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	How to mitigate potential attack vectors in a virtualized infrastructure?	L1	CO1	**02**
	b.	How does virtualization play a role in cloud infrastructure?	L1	CO2	**02**
	c.	Compare between private and public cloud.	L2	CO3	**02**
	d.	What are the Legal aspects impacting Cloud Security.	L1	CO4	**02**
	e.	Difference between Risk Avoidance and Risk Mitigation.	L1	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Illustrate in detail about how can hypervisor vulnerabilities be exploited by attackers, and what measures can be taken to mitigate these risks?	L1	CO1	**05**
		OR			
	b.	Explain in detail about protocols and encryption methods are recommended for securing communications between hosts and guests.	L3	CO1	**05**
3.	**a.**	Demonstrate in detail about to the key considerations when designing security controls for cloud infrastructure.	L1	CO2	**05**
		OR			
	b.	Discuss the common vulnerabilities found in cloud applications.	L2	CO2	**05**
4.	**a.**	Explain in detail about the key components of virtualization in cloud computing, and how do they interact.	L2	CO3	**05**
		OR			
	b.	Demonstrate in detail about role does encryption play in securing cloud applications, both at rest and in transit?	L1	CO3	**05**
5.	**a.**	Illustrate in detail about legal precedents exist concerning liability in cloud computing, and how do they influence current practices.	L1	CO4	**05**
		OR			
	b.	Explain in detail about measures can cloud providers take to ensure compliance with data protection laws while offering their services globally.	L1	CO4	**05**

6.	a.	Discuss in detail about how cloud computing can both mitigate and introduce risks to enterprise risk management. Provide examples for each case.	L1	CO5	**05**
		OR			
	b.	Describe the components of a cloud-specific risk management framework. How do these components differ from traditional IT risk management frameworks?	L1	CO5	**05**
7.	a.	Describe in detail about the common security challenges in securing communications between guest instances and the host.	L2	CO1	**05**
		OR			
	b.	Describe in detail about how data protection regulations like the GDPR impact the operations of cloud service providers, especially in terms of data storage and processing.	L2	CO4	**05**
		PART-C	**6 X 10=60**		
8.	a.	Discuss the best practices for securing data and applications in a PaaS environment.	L2	CO1	**10**
		OR			
	b.	Describe in detail about primary security concerns associated with hypervisors in a virtualized environment.	L2	CO1	**10**
9.	a.	Elaborate in detail about the primary risks associated with cloud infrastructure, and how can they impact an organization?	L2	CO2	**10**
		OR			
	b.	Outline the stages of a secure development life cycle (SDLC) for cloud applications.	L2	CO2	**10**
10.	a.	Evaluate the key elements of transparency in cloud computing, and how do they impact trust between service providers and consumers.	L4	CO3	**10**
		OR			
	b.	How does CloudTrust 2.0 integrate with existing security and compliance protocols in the cloud?	L3	CO3	**10**
11.	a.	Describe in detail about liability distributed between cloud service providers and customers in case of a data breach or service outage.	L2	CO4	**10**
		OR			
	b.	Illustrate in detail about the use of cloud services impact copyright ownership and enforcement, especially in relation to user-generated content.	L3	CO4	**10**

12.	**a.**	Outline the role of SLAs in managing performance and security risks in cloud computing. How should an SLA be structured to address these concerns?	L2	CO5	**10**
		OR			
	b.	How does cloud computing impact the traditional approaches to enterprise risk management (ERM)?	L2	CO5	**10**
13.	**a.**	Summarize the foundational principles of secure cloud application development.	L2	CO2	**10**
		OR			
	b.	Explain in detail about the main strategies for managing risks in the cloud environment, and how do they differ?	L3	CO5	**10**